NAME

mbgrd2gltf − convert bathymetric grid data to GLTF format with optional settings, including Draco compression.

SYNOPSIS

mbgrd2gltf −−input *FILE* [**−−binary**] [**−−output** *ROOT*] [**−−exaggeration** *NUM*] [**−−geoorigin**[=*LON,LAT,ELEV*]] [**−−draco**] [**−−quantize-position** *NUM*] [**−−quantize-normal** *NUM*] [**−−quantize-texcoord** *NUM*] [**−−quantize-color** *NUM*] [**−−html**] [**−−verbose**] [**−−help**]

Legacy single-dash syntax:
mbgrd2gltf −I*FILE* [**−B**] [**−O***ROOT*] [**−E***NUM*] [**−G**[*LON,LAT,ELEV*]] [**−D**] [**−Qp***NUM*] [**−Qn***NUM*] [**−Qt***NUM*] [**−Qc***NUM*] [**−W**] [**−V**] [**−H**]

DESCRIPTION

mbgrd2gltf is a tool designed to convert grid data files (GRD) to Graphics Library Transmission Format (GLTF) files. It can generate output in binary format to create a more compact GLTF file and exaggerate vertex altitude to enhance topographic features. The tool uses tiling to efficiently handle large datasets. Optional Draco compression can reduce the output GLTF file size significantly while maintaining good visual quality. With Draco, one can specify the quantization level, which affects the compression level of mesh vertices.

The output mesh vertices are positioned in an Earth-Centered, Earth-Fixed (ECEF) Cartesian coordinate system with units in meters. ECEF is a 3D right-handed coordinate system with its origin at Earth's center of mass, the Z-axis pointing toward the North Pole, the X-axis intersecting the equator at the prime meridian, and the Y-axis completing the right-handed system at 90 degrees east longitude.

For improved rendering precision in localized areas, the **−−geoorigin** option allows vertices to be expressed as offsets from a local origin point rather than from Earth's center. ECEF coordinates typically have magnitudes on the order of millions of meters, which can cause precision loss when using GPU single-precision floating point arithmetic. By specifying a GeoOrigin near the area of interest, coordinate magnitudes are reduced to thousands of meters, significantly improving rendering quality. The GeoOrigin can be explicitly specified with longitude, latitude, and elevation values, or automatically computed from the grid's geographic center and mean altitude. When no GeoOrigin is specified, vertices use the full ECEF coordinates.

MB-SYSTEM AUTHORSHIP

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MBGRD2GLTF AUTHORSHIP

The **mbgrd2gltf** program was contributed to MB-System by California State University Monterey Bay (CSUMB) Capstone teams. The original gltf-generator program was created for STOQS starting in Fall 2020:

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OPTIONS

−−input, −I
FILE
Input GMT GRD format bathymetry grid file. This option is required.

−−binary, −B
Generate the output in binary GLTF (GLB) format. Binary GLTF is more compact and typically loads faster in applications than text-based GLTF. Default: text-based GLTF output.

−−output, −O
ROOT
Specify the output file root. The appropriate file extension (.gltf or .glb) will be appended automatically. Default: input file basename in the same directory.

−−exaggeration, −E
NUM
Specify the vertical exaggeration factor as a decimal number, which multiplies the vertex altitudes. This can enhance the visibility of topographic features. Default: 1.0 (no exaggeration).

−−geoorigin, −G
[=*LON,LAT,ELEV*]
Specify a GeoOrigin point for high-precision local coordinates. All vertex coordinates will be expressed as offsets from this origin point in the ECEF coordinate system. This improves rendering precision for localized areas by reducing the magnitude of coordinate values, which helps overcome GPU single-precision floating point limitations.
With values (e.g., --geoorigin=-122.0,36.8,-1000): use specified GeoOrigin.
Without values: automatically compute and use grid center and mean altitude.
Not specified: vertices use original ECEF coordinates from Earth’s center (default).

−−draco, −D
Enable Draco mesh compression to reduce the file size by encoding meshes and point clouds. This can significantly reduce file sizes while maintaining visual quality.

−−quantize-position
NUM (or **-Qp***NUM*)
Draco position quantization bits (2-30). Affects the compression level of mesh vertex positions. Lower values increase compression but reduce precision, potentially affecting the visual quality of the terrain. Requires **−−draco** option. Default: 16.

−−quantize-normal
NUM (or **-Qn***NUM*)
Draco normal quantization bits (2-30). Affects the compression of surface normal vectors. Requires **−−draco** option. Default: 7.

−−quantize-texcoord
NUM (or **-Qt***NUM*)
Draco texture coordinate quantization bits (2-30). Affects the compression of texture coordinates.

Requires **−−draco** option. Default: 10.

−−quantize-color
NUM (or **-Qc***NUM*)
Draco color quantization bits (2-30). Affects the compression of vertex colors. Requires **−−draco** option. Default: 8.

−−html, −W
Generate an HTML viewer page with inline 3D visualization using X3DOM. The HTML file includes the glTF/GLB model with interactive navigation controls, processing provenance metadata (command line, timestamps, grid parameters, GeoOrigin settings, compression options), and links to MB-System and X3DOM documentation. The viewer requires serving the files via HTTP (e.g., python3 -m http.server) to avoid browser CORS restrictions when loading local files. Output file: <output_root>.html

−−verbose, −V
Enable verbose output showing NetCDF reading details and memory allocation information.

−−help, −H
Display help message and exit.

EXAMPLES

Convert a GMT GRD file to a GLTF file with default settings:
mbgrd2gltf --input input.grd

Convert a GRD file to GLB (Binary GLTF) with specified output folder and Draco compression:
mbgrd2gltf --input input.grd --binary --output /path/to/output/mymodel --draco --quantize-position 20

Convert with vertical exaggeration, Draco compression, and verbose output:
mbgrd2gltf --input input.grd --output /path/to/output/mymodel --draco --binary --quantize-position 20 --exagger

Convert bathymetry with 10x vertical exaggeration:
mbgrd2gltf --input bathymetry.grd --exaggeration 10.0 --binary

Convert bathymetry with a GeoOrigin for improved precision:
mbgrd2gltf --input monterey.grd --geoorigin=-122.0,36.8,-1000 --binary

Convert using automatic GeoOrigin (grid center):
mbgrd2gltf --input monterey.grd --geoorigin --binary

Generate HTML viewer with 3D visualization:
mbgrd2gltf --input bathymetry.grd --exaggeration 10 --html --binary

Legacy syntax (still supported):
mbgrd2gltf -I input.grd -B -O /path/to/output/mymodel -D -Qp 20 -E 10 -V

SEE ALSO

mbinfo(1), mbprocess(1), mblist(1)

BUGS

Please report any bugs on the MB-System GitHub Issue Tracker.